

tf.compat.v1.nn.depthwise_conv2d_native

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https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/depthwise_conv2d_native

Computes a 2-D depthwise convolution.

Function Signatures

```
tf.compat.v1.nn.depthwise_conv2d_native( input, filter,
strides, padding, data_format='NHWC', dilations=[1, 1, 1, 1],
name=None )
```

Code Examples

```
tf.compat.v1.nn.depthwise_conv2d_native(  
    input,  
    filter,  
    strides,  
    padding,  
    data_format='NHWC',  
    dilations=[1, 1, 1, 1],  
    name=None  
)
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    input,  
    filter,  
    strides,  
    padding,  
    data_format='NHWC',  
    dilations=[1, 1, 1, 1],  
    name=None  
)
```

```
for k in 0..in_channels-1  
    for q in 0..channel_multiplier-1  
        output[b, i, j, k * channel_multiplier + q] =  
            sum_{di, dj} input[b, strides[1] * i + di, strides[2] * j + dj, k]  
            *  
            filter[di, dj, k, q]
```

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for k in 0..in_channels-1  
    for q in 0..channel_multiplier-1  
        output[b, i, j, k * channel_multiplier + q] =  
            sum_{di, dj} input[b, strides[1] * i + di, strides[2] * j + dj, k]  
            *  
            filter[di, dj, k, q]
```

Documentation

Computes a 2-D depthwise convolution.

Given an input tensor of shape `[batch, in_height, in_width, in_channels]` and a filter / kernel tensor of shape `[filter_height, filter_width, in_channels, channel_multiplier]`, containing `in_channels` convolutional filters of depth 1, `depthwise_conv2d` applies a different filter to each input channel (expanding from 1 channel to `channel_multiplier` channels for each), then concatenates the results together. Thus, the output has `in_channels * channel_multiplier` channels.

Must have `strides[0] = strides[3] = 1`. For the most common case of the same horizontal and vertical strides, `strides = [1, stride, stride, 1]`.

Returns